9.1 Databases answers

Getting started

Complete the table for the following single-table database.

ID	Date	Time	Order Number
15	9/9/2020	15:00	2A
16	9/9/2020	15:05	2B
17	9/9/2020	15:11	2C
18	9/9/2020	15:15	2D
19	10/9/2020	10:00	3A
20	10/9/2020	10:30	3B

Number of fields	***4***
Number of records	***6***
The primary key	***ID***
Number of orders on 9/9/2020	***4***
Number of orders before 12:00	***2***
Data type for ID	***Integer or Text or Alphanumeric***
Data type for Time	***Date/Time or Time***
Data type for Order Number	***Text or Alphanumeric***

Practice

A job centre stores details about the jobs that are currently on offer. This information includes: a unique job number, the company the job is with, whether it is part or full time, the yearly salary.

Define a single-table database for the job centre by identifying the fields, data types, and primary key.

These are examples only, there are many possible correct answers:

Job Number	***Integer or Text or Alphanumeric***
Company	***Text or Alphanumeric***
Part Time	***Boolean***
Salary	***Real***

Primary key = Job Number

Challenge

Some databases need more than one table. Databases can also be set up to link tables together, so if you look at the data in one table it will align with the data in the other table. This is called a relationship.

Find an example where a database has more than one table, and where the tables are joined with a relationship. Identify the key components of the database tables, and how the relationship is set up between them.

Student answers will vary.

9.2 SQL answers

These question use the following database table, APPOINTMENT.

Appointment ID	CustomerID	Date	Time	Attended?	Fee
56	3DF	10/10/2020	09:30	Yes	125.00
57	6HF	10/10/2020	10:45	Yes	220.00
58	7VD	11/10/2020	10:00	No	100.00
59	2SD	11/10/2020	12:15	No	160.00
60	9JK	11/10/2020	15:00	Yes	200.00

Getting started

SQL query outputs:

```
SELECT CustomerID
FROM APPOINTMENT
```

```
SELECT AppointmentID
FROM APPOINTMENT
ORDER BY Time
```

```
SELECT CustomerID
FROM APPOINTMENT
WHERE Attended? = Yes
```

```
SELECT AppointmentID
FROM APPOINTMENT
WHERE Date = 10/10/2020
ORDER BY Time Desc
```

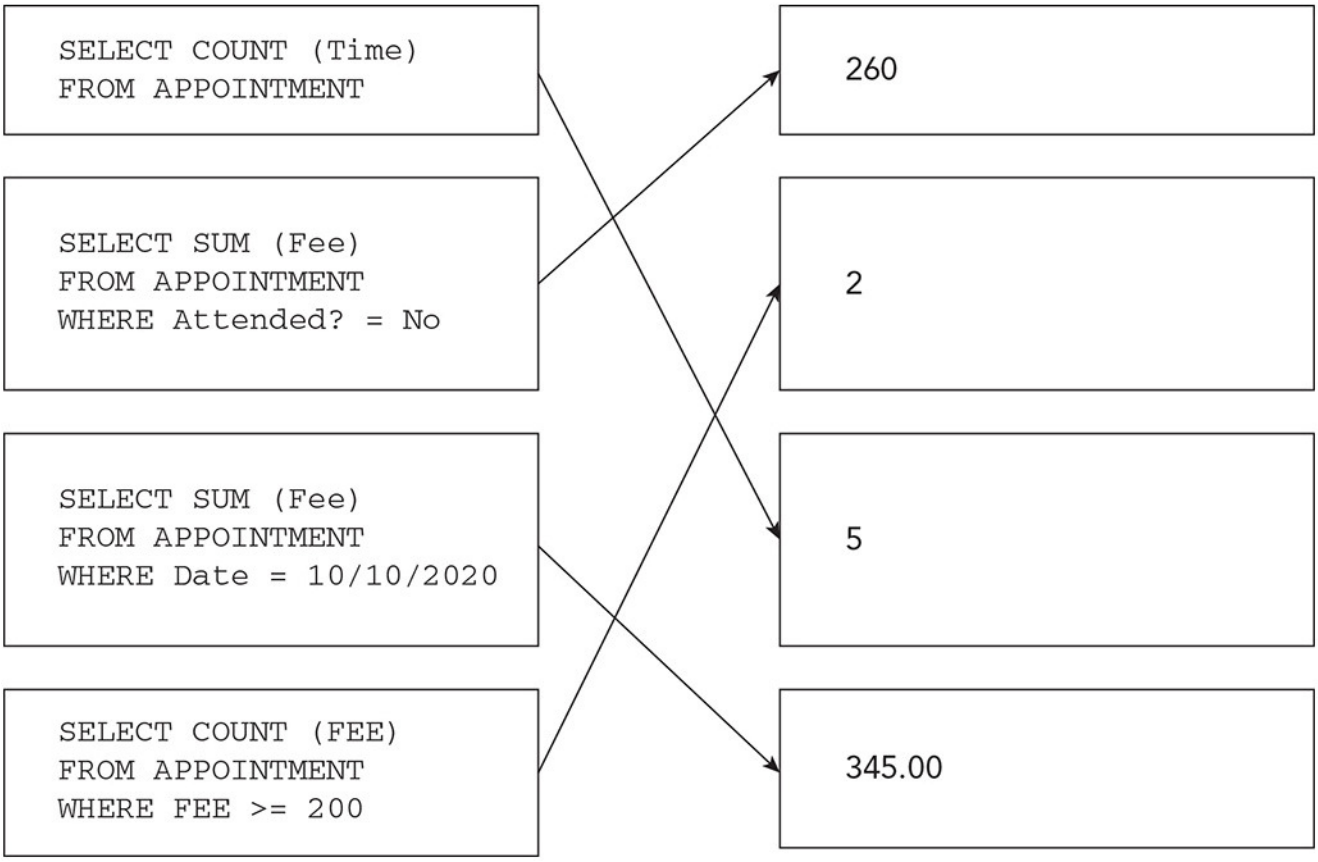
3DF
6HF
9JK

3DF
6HF
7VD
2SD
9JK

56
58
57
59
60

57
56

SQL query outputs:



Practice

Identify the output from the following SQL scripts:

1
```
SELECT Appointment ID, CustomerID
FROM APPOINTMENT
WHERE Date = 11/10/2020 AND Attended = No
```

58 7VD

59 2SD

2
```
SELECT Date, Time
FROM APPOINTMENT
WHERE Fee < 200
ORDER BY Fee DESC
```

11/10/2020	**12:15**
10/10/2020	**9:30**
11/10/2020	**10:00**

3
```
SELECT Appointment ID, CustomerID, Time, Attended?
FROM APPOINTMENT
WHERE Fee > 100 AND Fee < 200
```

56	**3DF**	**09:30**	**Yes**
59	**2SD**	**12:15**	**No**

4
```
SELECT Appointment ID, CustomerID
FROM APPOINTMENT
WHERE Date = 10/10/2020
ORDER BY Fee
```

56	**3DF**
57	**6HF**

5
```
SELECT COUNT(Appointment ID)
FROM APPOINTMENT
WHERE Attended? = No
```

2

6
```
SELECT SUM(Fee)
FROM APPOINTMENT
WHERE Time < 12:00
```

445.00

Challenge

Complete the following SQL scripts.

1 Return the Customer IDs of all customers attended an appointment on the 10/10/2020.

```
SELECT CustomerID
FROM APPOINTMENT
WHERE Date = 10/10/2020
```

2 Return the Appointment IDs of customer 7VD

```
SELECT Appointment ID

FROM APPOINTMENT

WHERE CustomerID = "7VD"
```

3 Return the Date and Time of all appointments on the 11/10/2020 where the fee is less than 100.00

```
SELECT Date, Time

FROM APPOINTMENT

WHERE Date = 11/10/2020 AND Fee < 100
```

4 Return the Fee of all customers who attended appointments on the 10/10/2020 in descending order of Time.

```
SELECT Fee

FROM APPOINTMENT

WHERE Date = 10/10/2020

ORDER BY Time DESC
```

5 Return the total cost of all appointments.

```
SELECT SUM(Fee)

FROM APPOINTMENT
```

6 Return the number of appointments between 10:00 and 13:00 inclusive.

```
SELECT COUNT (AppointmentID)

FROM APPOINTMENT

WHERE Time >= 10:00 AND Time <= 13:00
```

9.3 Database answers

Getting started

e.g. Show table

Field name	Data type	PK?	
date	***Date/time***	*Y*	
showName	***Text***		
type	***Text***		
length	***Integer***		

e.g. Booking table

Field name	Data type	PK?	
Booking ID	***Integer/Text***	*Y*	
Date	***Date/Time***		
NumberTickets	***Integer***		
BookingName	***Text***		
TelephoneNumber	***Text***		
Paid	***Boolean/Text***		

Practice

e.g. SQL queries

1. ***SELECT COUNT(BookingID) FROM Booking WHERE Date = #3/3/2021#***
2. ***SELECT TelephoneNumber FROM Booking WHERE Paid = False***
3. ***SELECT COUNT(date) FROM Show WHERE date >= #1/12/2021# and date <= #31/12/2021#***

Challenge

Identify **four** pieces of information you want to gather from the data.
Write an SQL script for each of these requirements.

There is no set answer as this depends on the learners and their choices.